

Personalised Learning Engagement And Recommendation System

(Interview Q/A)

1. How did dataset understanding influence feature selection in this project?

Understanding the dataset helped identify which features genuinely reflected student performance, such as academic records or engagement indicators. It prevented the use of irrelevant or misleading features and ensured meaningful inputs to the model.

2. What patterns did you discover during EDA?

EDA revealed variations in performance across different academic and behavioral attributes. It highlighted missing values, class imbalance, and correlations between features and outcomes, guiding preprocessing decisions.

3. Why is student data more sensitive than many other datasets?

Student data directly affects academic opportunities and personal development. Incorrect interpretation or misuse can impact confidence, evaluations, and future prospects. This makes careful handling and responsible use critical.

4. How would wrong predictions impact students or institutions?

Incorrect predictions could lead to unfair interventions, mislabeling, or missed support opportunities. Institutions might allocate resources inefficiently or create unintended bias.

5. What challenges arise when predicting academic performance?

Academic performance is influenced by many non-measurable factors such as motivation or environment. Data can be noisy, incomplete, or biased. This makes precise prediction inherently difficult.

6. How did preprocessing choices affect fairness?

Proper preprocessing reduced unintended bias by handling missing values and encoding categories consistently. Poor preprocessing could disproportionately affect certain student groups, leading to unfair predictions.

7. How would you explain model results to a teacher?

I would explain which factors most influenced predictions, such as attendance or prior scores. The focus would be on trends and indicators rather than deterministic outcomes. The model acts as a support tool, not a judgment.

8. What limitations does this student prediction model have?

The model relies only on historical data and cannot capture real-time changes. It may not generalize across different institutions or education systems. Predictions should be interpreted cautiously.

9. How would you extend this project to improve reliability?

I would incorporate longitudinal data, more behavioral indicators, and cross-validation across different cohorts. Regular retraining would help keep predictions relevant.

10. What did this project teach you about real-world ML problems?

It highlighted that data quality, context, and responsible interpretation matter more than model complexity. Real-world ML requires balancing performance, explainability, and ethical considerations.
